6.10 Integration of Rational Functions Using Partial Fractions

Name _____ Period ____ Date _____

AP Calculus 6.10 · Student sheet

I can…

- write a partial fraction decomposition with distinct linear factors.
- find the constants A and B.
- integrate using partial fractions.

Vocabulary · fill in the missing word

Partial fraction decomposition — writing a _____ function as a sum of simpler fractions, like $\frac{A}{x-a}+\frac{B}{x-b}$

Proper rational function — the numerator has a smaller degree than the _____

Part 1 · Add, then undo

Add, then reverse.

Step	Result
$1/(x-1)-1/(x+1)$	
$\int$ of the pieces	

Part 2 · Notes and examples (fill in each note, then work the example)

Distinct linear factors: $(px+q)/((x-a)(x-b)) = A/(x-a) + $ _____.

Finding A and B: multiply through, then substitute the _____ of the factors.

Example 1

Decompose $\frac{5x-1}{(x-1)(x+1)}$.

Integrate: each piece gives a _____: $\int A/(x-a)\,dx = A\ln|x-a| + C$.

Example 2

Find $\int \frac{5x-1}{(x-1)(x+1)}\,dx$.

Example 3

Find $\int dx/(x^2-4)$.

Example 4

Evaluate $\int_1^2 dx/(x(x+1))$.

Part 3 · Partner practice

Partner 1

Decompose $\dfrac{1}{x(x+2)}$.

Partner 2

(AP® Practice) $\int \dfrac{3}{(x-2)(x+1)}\,dx =$

(A) $\ln|x-2| - \ln|x+1| + C$ (B) $\ln|x-2| + \ln|x+1| + C$

(C) $3\ln|(x-2)(x+1)| + C$ (D) $\ln|x+1| - \ln|x-2| + C$